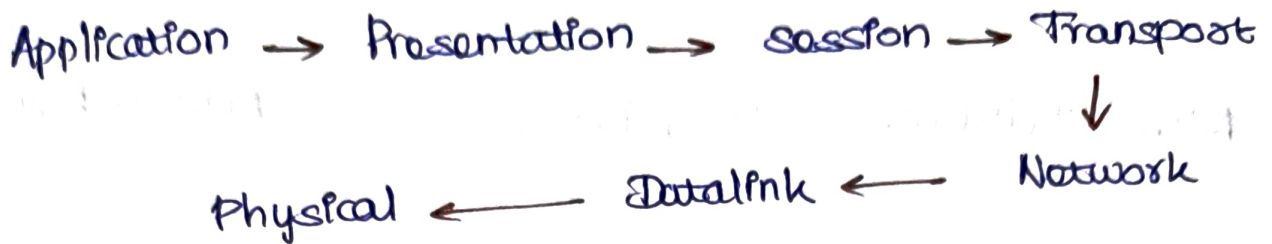


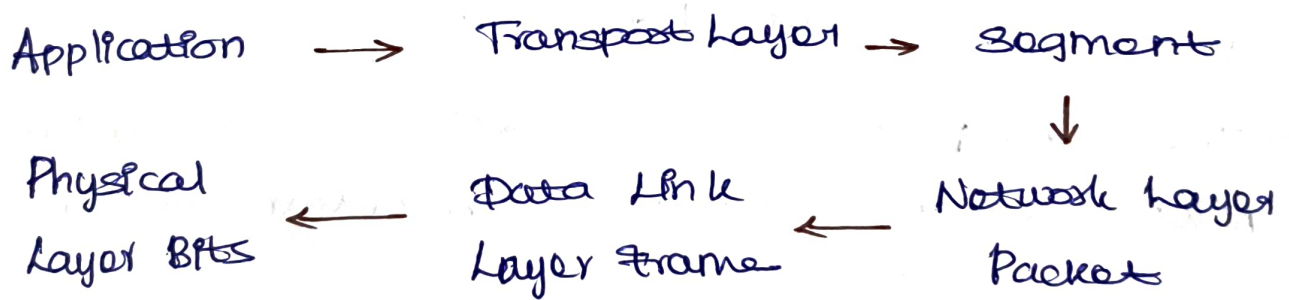
1. Missing OSI Layers



Analysis

- Session Layer establishes, manages and terminates communication session between devices
- Data Link layer provides framing, MAC addressing, error detection, and reliable node-to-node delivery
- Together, these layers ensure communication is synchronized, error-free and reliable from source to destination.

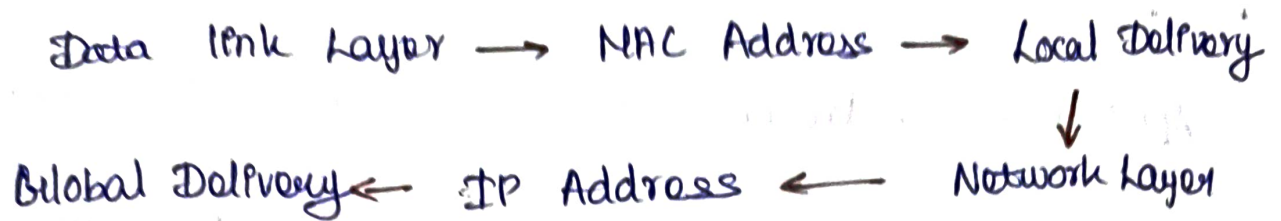
2. Data Units and OSI Layers



Analysis

- Transport Layer divides data into segments
- Network Layer adds IP information to form packets
- Data Link Layer converts frames into bits for transmission.

3. MAC Address and IP Address



Analysis

- The MAC address uniquely identifies devices within the same local network (LAN)
- The IP address identifies devices across different networks and enables routing over the Internet
- Together, MAC and IP addressing ensure complete and accurate data delivery from source to destination.

4. Error Control - Analysis:

- The Data Link Layer detects transmission errors using CRC / checksum
- The transport layer corrects errors using ACKs, sequence numbers, and retransmission
- Together, they ensure reliable error-free communications.

5. Web Browsing Using OSI Layers - Analysis

- Each OSI layer performs a specific function during web browsing.
- If one layer fails, the upper layers cannot communicate.

→ Hence, successful communication depends on all layers working together.

6. Network Troubleshooting - Analysis

→ The physical Layer and Data Link Layer are functionally correctly

→ The failure starts at the Network Layer, indicating an IP addressing.

→ The concept map helps identify the exact layer where the fault occurs.